

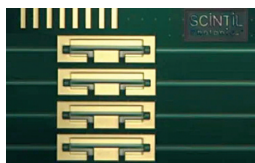
include bigger batteries and offer longer run times without affecting the sound quality. Moreover, the MEMS speakers feature a wide bandwidth, fulfilling hi-res audio up to a bandwidth of 40kHz, thus enabling them to produce a natural and clear sound. The USound MEMS speakers don't use any ferromagnetic materials, making them suitable for developing MRI-compatible headphones. The Kore 4.0 speakers are suitable for use in high-quality headphones, hearing aids, AR/VR glasses, etc.

USound

<https://usound.com>

DFB comb laser source

SCINTIL from Scintil Photonics is a single-chip multi-port 100GHz DFB Comb Laser Source that significantly enhances computing system performance and power efficiency in high-performance computing and AI applications. It is the first fully integrated single chip



that achieves 100GHz frequency spacing, offering twice the number

of optical carriers compared to existing solutions and enabling doubled transmission speed. The SCINTIL Comb Laser Source allows for uncooled dense wavelength division multiplexing (DWDM) links with twice the density of optical carriers. It supports configurations for transmitting 16 optical carriers at 64Gbps per fibre, making it ideal for next-gen optical compute interconnect links. The solution offers very narrow controlled channel spacing, which can enable increasing the number of optical carriers in a single fibre. SCINTIL's technology can enable engineers to offer better efficiency and performance in interconnects for high-performance computing and AI applications.

Scintil-Photonics

<https://www.scintil-photonics.com>

Graphene based current sensor

The compact Paragraf graphene based current sensor can measure the current flow direction and magnitude of the

current. It allows for the measurement of the magnetic fields generated by the flow of current, without being part of the circuit. This feature makes it possible for designers to integrate the sensor



without modifying the circuit, thus simplifying the process while also reducing the

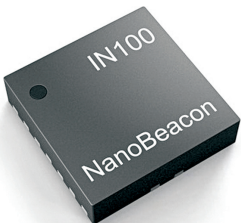
cost and time of development. The sensor only responds to magnetic fields perpendicular to the sensing layer, which provides a much more accurate field and current mapping for battery applications. Paragraf's current sensors enable cell-level monitoring, and the 2D geometry of the graphene based sensors allows for high immunity to stray magnetic fields in the plane of the sensor, which is critical in electrically noisy environments. The sensor is suitable for use in battery packs to mitigate thermal runaways and fire accidents. The sensor can be employed in larger battery packs, where the current sensor's low power requirement and the ability to place the sensor externally to the cell circuit will enable complete cell-level mapping of the battery, thus improving the safety and reliability of battery packs.

Paragraf

<https://paragraf.com>

Bluetooth SoC

The IN100 NanoBeacon SoC from InPlay is the smallest Bluetooth SoC measuring just 1.1mm × 2.0mm × 0.35mm. It integrates Bluetooth 5.3 technology, features a firmware-less



design, and employs a unique protocol implemented directly into its circuitry. The NanoBeacon supports

I2C, ADC, 1-wire, and GPIO interfaces, making it compatible with common sensors. The device enables seamless real-time wireless communication over Bluetooth and operates in the 2.4GHz RF frequency band. It offers advantages

such as low cost, low power consumption, and enhanced security. With sub-microwatt power consumption, it can run for years on a tiny battery. The NanoBeacon has 4k SRAM + 4k OTP memory, comes in DFN8 and QFN18 packages, and operates from -40 to +125°C. The IN100 NanoBeacon SoC is suitable for a wide variety of Industrial Internet of Things (IIoT) applications, including asset management, active RFID, smart sensors, smart healthcare, etc. It will enable engineers to develop low-cost, low-power consumption, firmware-less design, making it a versatile solution for various IIoT applications.

Inplay Tech

<https://inplay-tech.com>

Human presence sensor

Ai-Thinker's Rd-01 24G radar module is one of the smallest human body presence sensing modules available in the market. It is a three-in-one module which employs Wi-Fi, BLE, and radar for communication for use in a wide variety of areas, such as smart hotels, smart homes, office buildings, etc. This



module is equipped with the BL602 and S3KM111L chips that can accurately detect human body presence, including motion, micro-movement, and stillness. Its integration of radar sensing technology into the hospitality industry offers a unique advantage. By deploying this module in buildings, factories, hotels, etc, the operators can improve energy consumption costs, check-in experiences, occupancy rates, and flight risk management. The combination of Wi-Fi, BLE, and radar in a single module provides a comprehensive solution for presence sensing and monitoring. With its reliable performance and precise insights into human presence and behaviour, the Rd-01 24G radar module can enhance overall efficiency and human satisfaction in deployed areas.

Ai-Thinker

<http://en.ai-thinker.com>